Subject-ELE-I-Cloud, Microservices and Applications Semester- Vs	
Sr. No.	**Description of Question**
Unit-I	
1	Explain Cloud Computing Architecture.
2	What are the advantages and disadvantages of Cloud Computing?
3	Discuss how Cloud Computing is used in Education/ Medical/ Manufacturing with the help of suitable diagram.
4	What is the importance of cloud computing in a modern enterprise's IT strategy?
5	Explain the difference between IaaS, PaaS, and SaaS.
6	What is meant by cloud deployment models? Provide an example.
7	Compare and contrast the advantages and disadvantages of using a public cloud versus a private cloud.
8	Explain service-oriented architecture (SOA). Discuss how it can benefit cloud computing applications with the help of suitable diagram.
Unit-II	
1	Explain HDFS Structure with all the components.
2	How do scalability and elasticity differ in cloud computing, and why are both important?
3	What are the advantages and disadvantages of Type 1 and Type 2 hypervisors in a cloud environment?
4	Explain the overall process of MapReduce and represent MapReduce word count for the input given **(Input will be given with the question)**
5	Describe algorithms used for load balancing with the help of diagrams.
6	Differentiate between the benefits and challenges of using virtualization technologies.
7	Explain the role of Hypervisor. Explain working of Type1 and Type 2 hypervisors with the help of suitable diagram.
8	Why is disaster recovery important? Describe the key elements of a disaster recovery plan?
Unit-III	
1	Describe IAM components. Enlist features of IAM.
2	With the help of suitable diagram explain a CSA Cloud Security Architecture and it working.

3	In what way would you utilize cloud security monitoring tools to protect that data? Explain with the help of suitable scenario.
4	**Explain** three key benefits of implementing cloud security measures.
5	**Describe** the authentication process in cloud environments and **explain** how it contributes to security.
6	**Explain** the role of encryption in cloud security and **discuss** how it helps in protecting data at rest and in transit.
7	**Differentiate** between authentication and authorization in the context of cloud security
8	**List** common risks associated with cloud security and **discuss** the challenges organizations face in mitigating these risks.
Unit-IV	
1	Describe are the 12 factors of a cloud-native application? why are they important in modern software development?
2	What are the advantages and disadvantages of using a monolithic architecture compared to microservice architecture?
3	Explain the fundamental principles of microservice design.
4	How does Spring Boot simplify the development of microservices?
5	Explain the application integration process and its significance in Microservices architecture.
6	Explain the design approach for creating microservices and how it differs from traditional development methods.
7	Compare monolithic and distributed architectures.
8	Identify common tools used for API development and explain their functionalities.
9	Define what an API is and explain its role in software development.
10	Discuss the benefits of Microservices architecture
Unit-5	
1	Explain the Docker architecture. What are the steps involved in the containerization process.
2	Explain the working of two commonly used DevOps tools in cloud application development.
3	Describe the DevOps Lifecycle.
4	What are the benefits of using Docker for cloud-native applications compared to traditional virtual machines?
5	Explain how integrating DevOps practices with Docker can enhance the efficiency of

	cloud application development.
6	Explain the core principles of DevOps and describe how they improve the software delivery lifecycle.
7	Explain three popular tools used in cloud application development and their primary functionalities.
8	What is API? Explain its role in software development.
9	State the difference between Virtual Machine and Containers.
10	Demonstrate how Continuous Integration (CI) tools contribute to maintaining code quality and deployment speed.
11	Explain the benefits of containerization over traditional virtualization methods in application deployment.

Prof. R. M. Shete
Subject In-Charge